

Assignment Overview

This assignment invites you to demonstrate your ability to critically engage with one of the seminar topics covered during the semester while also reflecting on the evolving role of Generative AI and LLMS in academic work. You will choose **one topic** from the module and complete the assignment using **one of three AI integration approaches** outlined below.

The purpose of this assessment is twofold:

1. To deepen your understanding of computer science concepts through independent research and critical writing.
 2. To develop your ability to evaluate the strengths, limitations, and implications of using Generative AI in academic and professional contexts.
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Step 1: Topic Choice

Choose one of the topics covered in the module seminars (e.g., What is AI, Ethics of AI, Surprise in AI, Running with Data, Time series classification etc.).

Step 2: Step 2: LLM Integration Choice

Choice 1: No Gen AI

- Construct your essay entirely through self-directed research without the assistance of Generative AI tools.
- Your essay should critically explore your chosen topic, supported by relevant academic sources.
- **Key Benefit:** Develops traditional academic research and critical analysis skills.

Why choose this approach

- You wish to maintain complete authorial control and voice.
- You prefer to rely entirely on self-directed research and reflection.
- You want to focus only on computational thinking concepts, without AI support.

Process

Create an essay with the following structure:

1. Abstract (250 words)

- Clearly state your chosen topic.
- Summarise your planned argument or line of inquiry.

2. Critical Reflection on the Topic (1,400 words)

- Articulate your chosen computational thinking topic and why it is significant.
- Critically analyse key authors' views and debates surrounding this topic.

3. Impact of Generative AI/LLMs (400 words)

- Critically reflect on how what your experience of writing the assignment was like without the use of LLM tools.

4. References

- Include 3 – 5 academic references.
- Use consistent referencing style

Choice 2: Using an AI tool for Task Completion, Human Evaluation

Why choose this approach

- You want to experience GenAI's strengths and limitations in generating academic writing.
- You are interested in critically evaluating GenAI outputs against academic standards.
- You wish to better understand issues of academic integrity and AI literacy.

Process

1. Generate AI Content

- Prompt a GenAI tool (e.g., ChatGPT) to produce a 2,000-word essay (with 3 -5 10 references) on your chosen topic, it must also include a 250 word abstract.

2. Critically Reflect on the Evidence (1,400 words)

- Compare and critique the AI-generated essay alongside your own analysis of the topic.

- Use the **UBC Evaluating AI Text Output Rubric** (Appendix 1) to guide your evaluation of the AI's accuracy, relevance, clarity, completeness, contextual understanding, creativity, bias, and sources.
- 3. **Experience of Using GenAI (400 words)**
 - Identify which tool you used, describe your process, and reflect on surprises or limitations.
 - Critically evaluate the experience of working with AI.
- 4. **References**
 - Include 3 - 5 academic references
 - Use consistent referencing style
- 5. **Appendix (not included in word count)**
 - Include all prompts and outputs used, in chronological order.

Choice 3: GenAI for Idea Generation, Structuring, and Editing

- Use a Generative AI tool throughout your writing process—for brainstorming, planning, structuring arguments, or editing drafts.
- **Key Benefit:** Builds practical AI literacy while retaining ownership of your academic work.

Why choose this approach

- You want to explore GenAI as a thinking partner rather than a text generator.
- You wish to develop skills in AI-supported writing processes such as brainstorming, structuring, or editing.
- You are interested in reflecting on how GenAI reshapes academic practices and skills.

Process

Use GenAI iteratively during your writing process for support such as:

- Generating outlines for your essay.
- Suggesting alternative perspectives or counterarguments.
- Checking your writing for clarity, coherence, or bias.
- Exploring theoretical frameworks or summarising debates.

Create an essay with the following structure:

1. **Abstract**
 - Clearly state your chosen topic.
 - Summarise your planned argument or line of inquiry.
2. **Critical Reflection on the Evidence (1,400 words)**

- Articulate your chosen computational thinking topic
 - Critically analyse key authors' views and debates surrounding this topic.
- 3. Experience of Using GenAI (400 words)**
- Describe when and how you used GenAI.
 - Reflect on how the AI-supported process influenced your thinking.
 - Critically evaluate the benefits and drawbacks.
- 4. References**
- Include 3- 5 academic references,
 - Properly cite any GenAI tools used.
- 5. Appendix (not included in word count)**
- Include all prompts and AI responses you engaged with, labelled and organised.

Assessment Criteria

Your submission will be assessed on:

- **Engagement with Content (30%):** Depth of understanding and critical engagement with the chosen topic.
- **Research and Evidence (20%):** Use of appropriate academic sources, correct referencing, and integration of evidence into your argument.
- **Critical Engagement with AI (20%):** Quality of reflection on AI's role, limitations, and implications when conducting research and synthesising knowledge.
- **Clarity and Structure (15%):** Organisation, coherence, and clarity of writing.
- **Reflection and Self-Assessment (15%):** Depth of insight into your own learning process and/or your interaction with AI.

Submission Requirements

- Word count: 2,000 words ($\pm 10\%$).
- Format: Word or PDF document, double-spaced, 12pt font.
- References: Consistent referencing style, 3-5 academic sources.
- For Choices 2 and 3: Include your AI prompts and outputs in an appendix (not included in the word count).

Academic Integrity

This assignment is designed to support **responsible AI use**. You must clearly document how you used (or chose not to use) Generative AI. Misrepresentation of AI use will be considered a breach of academic integrity.

Criteria	Excellent	Good	Satisfactory	Needs Improvement
1. Critical Engagement with Topic (40%)	Demonstrates outstanding understanding of the chosen topic. Argument is insightful, original, and strongly supported by relevant literature.	Demonstrates clear understanding with well-supported arguments. Some originality evident.	Demonstrates adequate understanding with limited depth or originality. Some key ideas underdeveloped.	Demonstrates poor or inaccurate understanding of the topic. Lacks critical analysis and originality.
2. Research and Evidence (25%)	Uses a wide range of high-quality, relevant academic sources (±5). Evidence is seamlessly integrated into arguments. Correct referencing throughout.	Uses a sufficient range of relevant academic sources. Some integration into arguments. Minor referencing errors.	Uses limited or partially relevant sources. Evidence sometimes disconnected from argument. Frequent referencing errors.	Uses few or no relevant academic sources. Evidence absent, poorly integrated, or inappropriate. Referencing inconsistent or missing.
3. Critical Reflection on GenAI (20%)	Provides a sophisticated, critical reflection on AI as a tool. Shows excellent awareness of AI's potential, limitations, and ethical issues. Reflection is deeply personal and analytical.	Provides a clear reflection with good awareness of AI tools' potential and limitations. Some personal insights.	Provides a basic reflection with limited critical analysis. Discussion is descriptive rather than analytical.	Provides little or no reflection. Lacks awareness of AI's role, limitations, or implications.
4. Structure, Clarity and Academic Writing (15%)	Exceptionally clear, coherent, and well-structured writing. Arguments flow logically. Excellent academic style.	Clear and well-structured writing with logical progression of ideas. Good academic style.	Writing is understandable but may lack coherence or logical flow. Style inconsistent.	Writing is unclear, poorly structured, or significantly impedes understanding.

UBC Evaluating AI Text Output: Rubric (adapted)

Accuracy	Assess the accuracy and reliability of the AI response and determine whether the information provided is correct and/or factual information.
Relevance	Evaluate the extent to which the AI response directly addresses the main points or concerns raised in the question or prompt.
Clarity and Coherence	Consider the clarity, organization, and coherence of the AI response to convey information effectively.
Completeness	Evaluate whether the AI response covers all essential aspects or components of the topic, and whether it omits crucial details.
Contextual Understanding	Assess the AI's ability to grasp the context of the question or prompt and whether it provides a response that demonstrates understanding of the specific situation or background information.
Creativity and Originality	Assess the AI response for any unique or creative elements. Does it provide novel insights or alternative perspectives? Does it go beyond generic or predictable responses?
Bias	Check for diversity of sources to address ableist thinking, gender bias, and racial bias. Look for emotionally charged language, exaggerations, or generalizations that sway opinion. Pay attention to what is included and omitted in the information.
Sources and Citations	Check the sources and citations used in the content. Assess whether they are credible, reputable, unbiased, and represent a diversity of viewpoints.